

PSI

Center for Scientific Computing,
Theory and Data

Koopmans functionals in practice

minimisation, screening coefficients, automation, and more...

Edward Linscott

ICTP–MARVEL College, 27 May 2026

$$E_{\text{Koopmans}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(\underbrace{- \int_0^{f_i} \varepsilon_i(f) \, df}_{\text{removes curvature}} + \underbrace{f_i \eta_i}_{\text{restores linearity}} \right)$$

¹I. Dabo *et al.* *Phys. Rev. B* **82**, 115121 (2010), G. Borghi *et al.* *Phys. Rev. B* **90**, 75135 (2014), N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

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- orbital-density dependence
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Differences to semi-local functionals:

- different flavours
- orbital-density dependence
- screening

1

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Flavours of Koopmans functionals

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One degree of freedom: what should be the gradient of this linear term?

-

$$E_{\text{KI}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \int_0^1 \varepsilon_i(f) \, df \right)$$

One degree of freedom: what should be the gradient of this linear term?

- the base functional $\rightarrow$ “KI” (Koopmans integral). Enforces IP theorem. Does not affect energy/density!
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$$E_{\text{KIPZ}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \left\{ \int_0^1 \varepsilon_i(f) \, df - E_{\text{Hxc}}[n_i] \right\} \right)$$

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- the base functional $\rightarrow$ “KI” (Koopmans integral). Enforces IP theorem. Does not affect energy/density!
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You might also see...

- “pKIPZ” = KIPZ Hamiltonian evaluated on the KI solution
- “K” = an earlier iteration based off half-filling rather than integer endpoints (no longer used)

Orbital-density-dependence



$$-\int_0^{f_i} \varepsilon_i(f) \, df + f_i \int_0^1 \varepsilon_i(f) \, df = E_{\text{Hxc}}[\rho] + E_{\text{Hxc}}[\rho - \rho_i] + f_i(-E_{\text{Hxc}}[\rho - \rho_i] + E_{\text{Hxc}}[\rho - \rho_i + n_i])$$

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For filled orbitals with KI:

$$v_i^{\text{KI}}/\alpha_i = -E_{\text{H}}[n_i] + E_{\text{xc}}[\rho] - E_{\text{xc}}[\rho - n_i] - \int d\mathbf{r}' v_{\text{xc}}(\mathbf{r}', [\rho]) n_i(\mathbf{r}')$$

Orbital-density-dependence



2

¹G. Borghi *et al.* *Phys. Rev. B* **91**, 155112 (2015)

²N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018)

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Orbital-density-dependence

Initialise with MLWFs, then (optionally) solve with CG minimisation:

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For more details see¹

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Gives rise to a set of minimising orbitals (localised/variational)

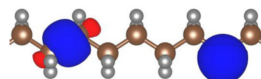


Figure 1: variational

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¹G. Borghi *et al.* *Phys. Rev. B* **91**, 155112 (2015)

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Gives rise to a set of minimising orbitals (localised/variational)

Diagonalising at the minimum gives rise to diagonalising orbitals (delocalised/canonical)

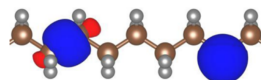


Figure 1: variational

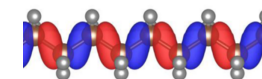


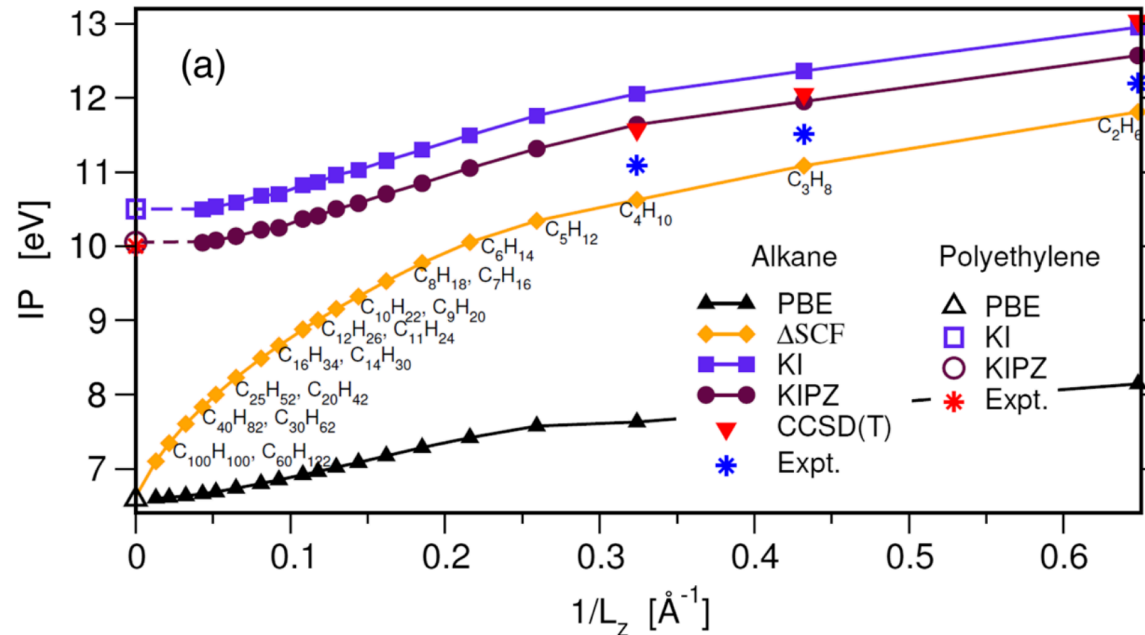
Figure 2: canonical

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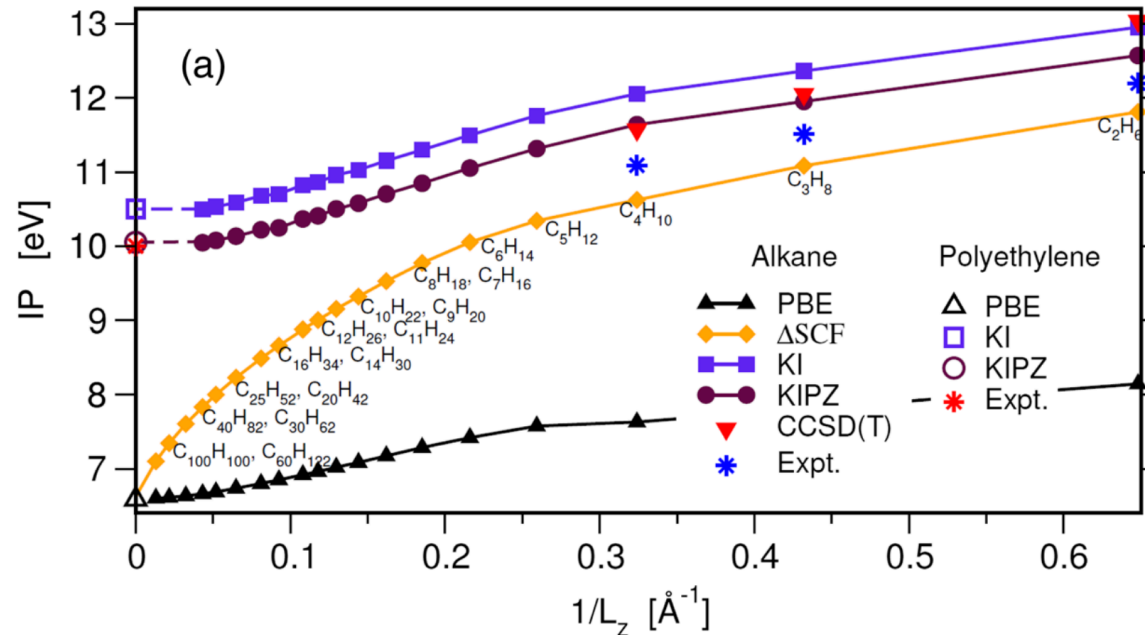
Importance of localisation



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¹N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018)

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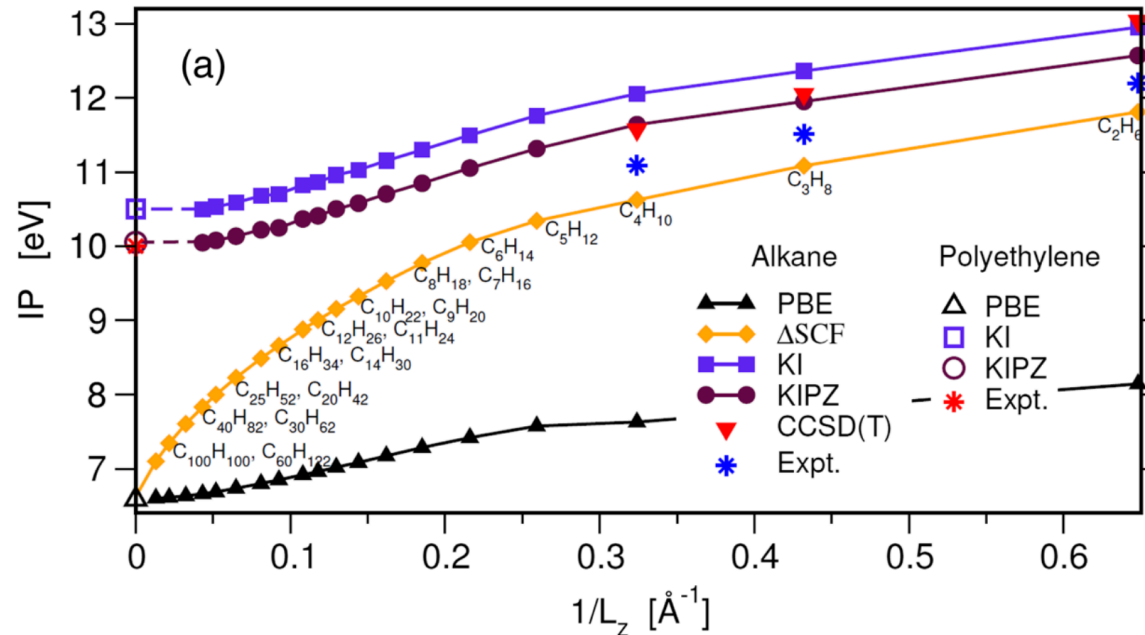


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In the bulk limit for one cell $\Delta E = E(N - \delta N) - E(N)$

¹N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018)

Importance of localisation



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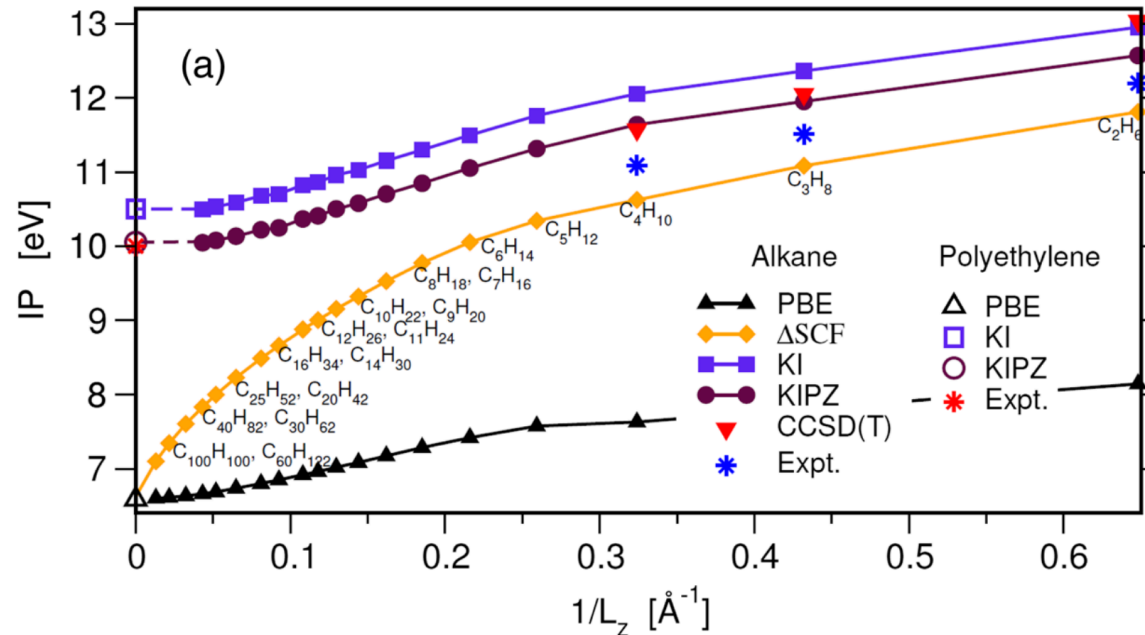
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Importance of localisation

Across all the cells $\Delta E = \frac{1}{\delta N} (E(N - \delta N) - E(N)) = -\frac{dE}{dN} = -\varepsilon_{\text{HO}}$

Importance of localisation



1

$$\lim_{n_i(r) \rightarrow 0} v_i^{\text{KI}} / \alpha_i = \lim_{n_i(r) \rightarrow 0} -E_{\text{H}}[n_i] + E_{\text{xc}}[\rho] - E_{\text{xc}}[\rho - n_i] - \int d\mathbf{r}' v_{\text{xc}}(\mathbf{r}', [\rho]) n_i(\mathbf{r}') = 0$$

¹N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018)

Orbital-density-dependence

Other features of orbital-density-dependence

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¹A. Ferretti *et al.* *Phys. Rev. B* **89**, 195134 (2014)

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Other features of orbital-density-dependence

- ODD functional means that we know $\hat{H}|\varphi_i\rangle$ for variational orbitals $\{|\varphi_i\rangle\}$ but we don't know $\hat{H}$ in general
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- Practically we can often use MLWFs
- a natural generalisation in the direction of spectral functional theory (as discussed already by Andrea)¹

¹A. Ferretti *et al.* *Phys. Rev. B* **89**, 195134 (2014)

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¹C. Li *et al. J. Chem. Phys.* **146**, 74107 (2017)

- In Hartree-Fock (the original “Koopmans’ theorem”):

$$E_{\text{ee}}^{\text{HF}} = \frac{1}{2} \sum_{ij} f_i f_j \int d\mathbf{r} d\mathbf{r}' \frac{|\psi_i(\mathbf{r})|^2 |\psi_j(\mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} - \frac{\psi_i^*(\mathbf{r}) \psi_j^*(\mathbf{r}') \psi_i(\mathbf{r}') \psi_j(\mathbf{r})}{|\mathbf{r} - \mathbf{r}'|}$$

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- Account for screening post-hoc:

$$\frac{dE}{df_i} \approx \alpha_i \frac{\partial E}{\partial f_i}$$

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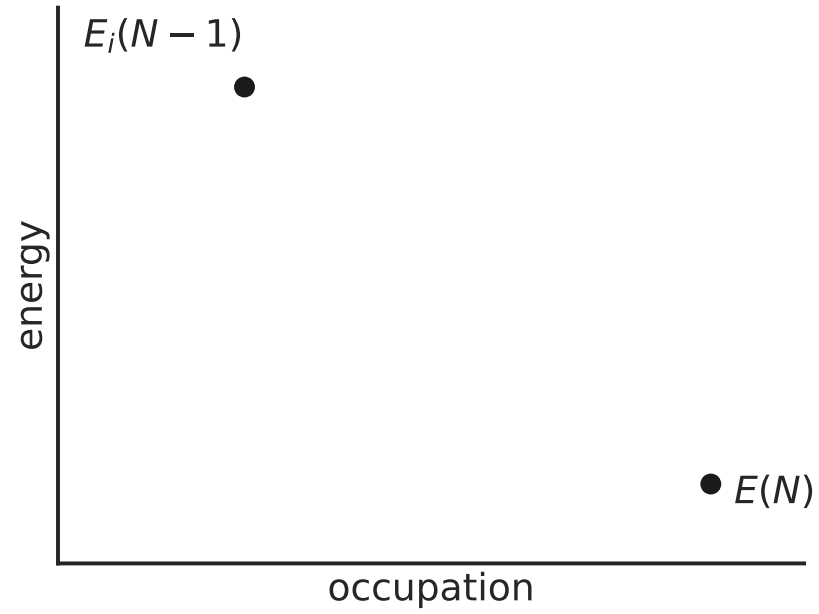
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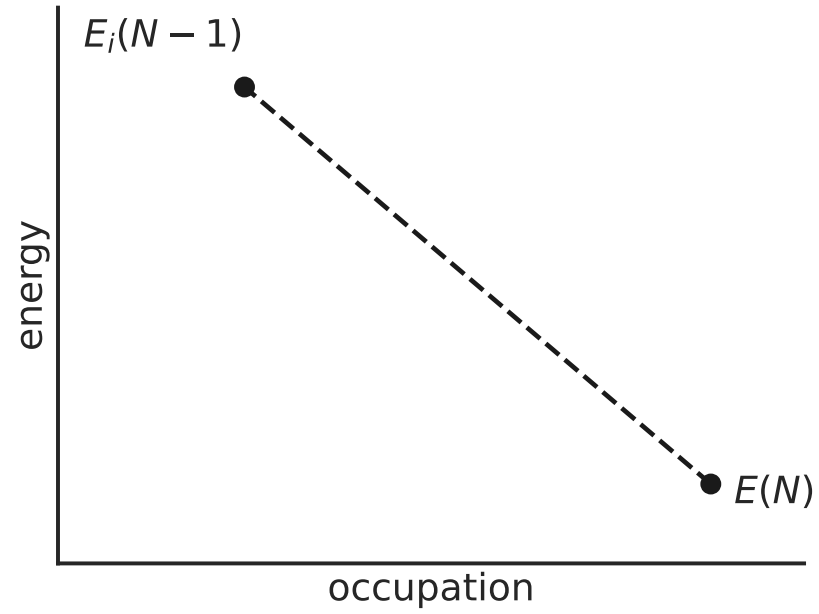
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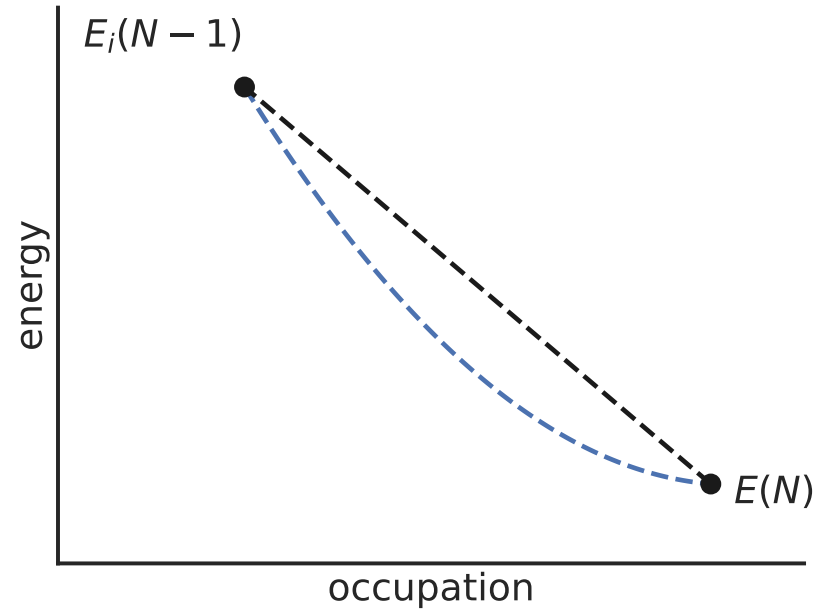
- How to choose an appropriate value for α_i ? Return to the original idea of Koopmans functionals:

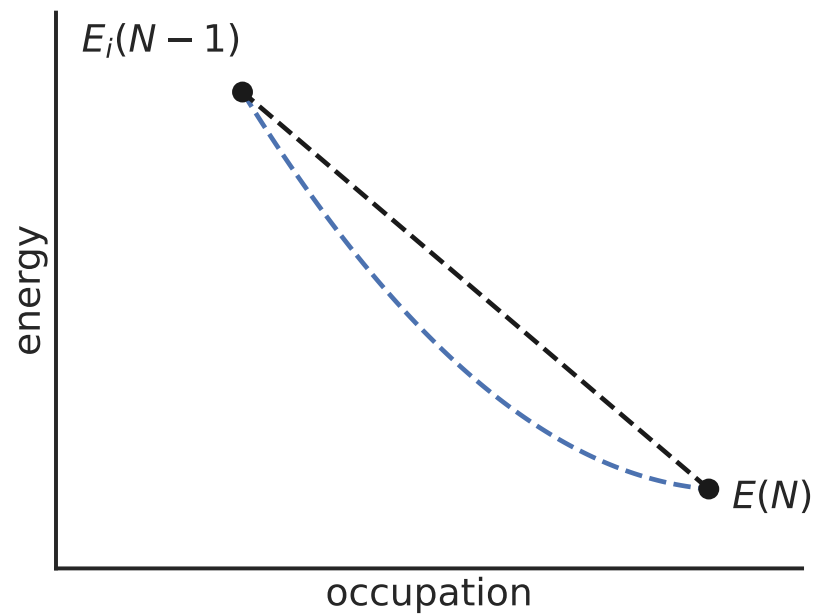
$$\varepsilon_i^{\text{Koopmans}} = E_i(N - 1) - E(N)$$

¹C. Li *et al.* *J. Chem. Phys.* **146**, 74107 (2017)

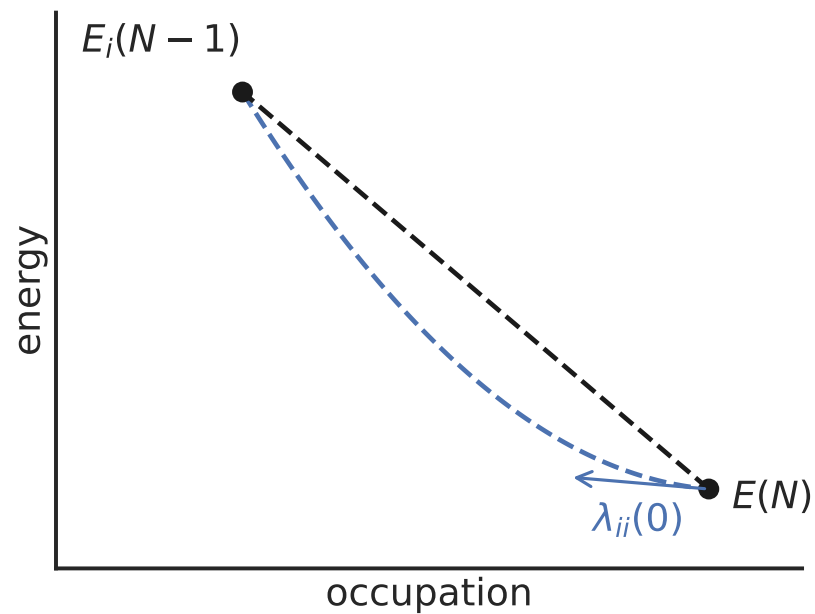




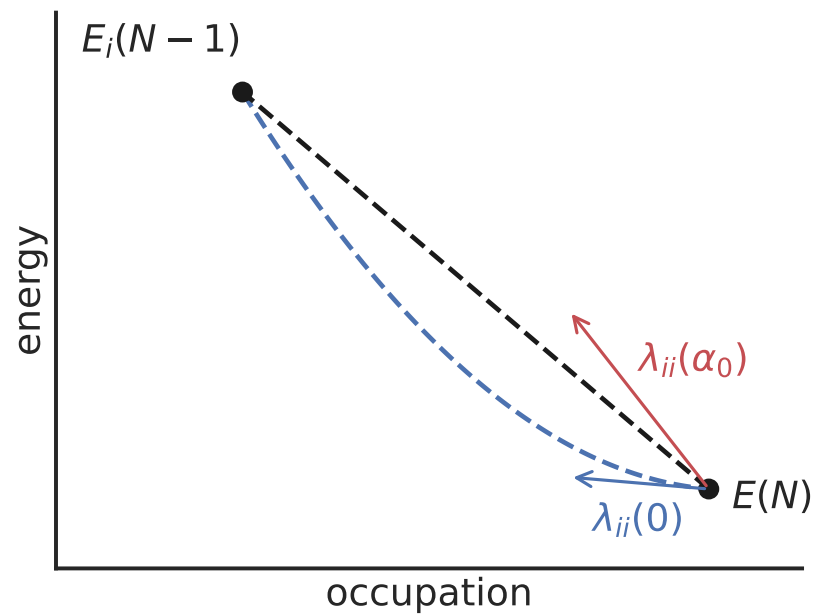




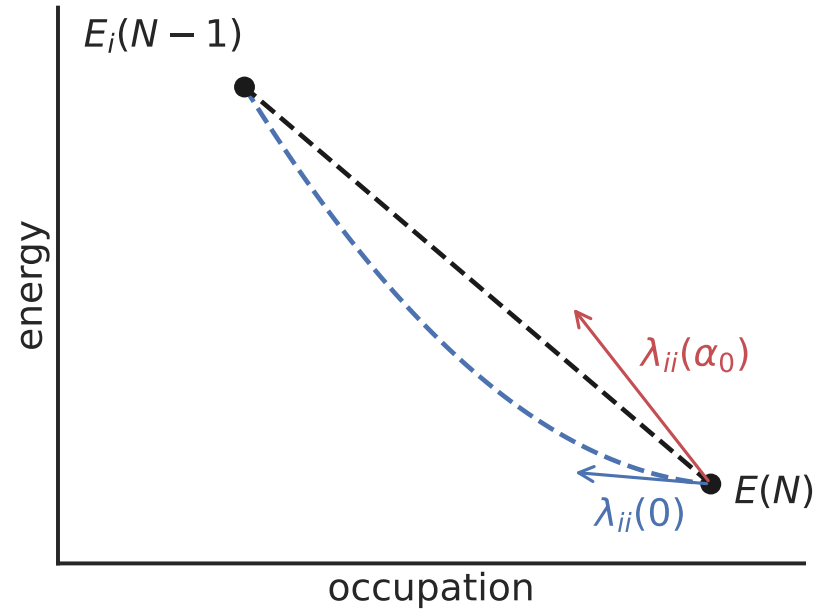
$$\lambda_{ii}(\alpha) \equiv \langle \varphi_i | \hat{h}^{\text{DFT}} + \alpha \hat{v}^{\text{Koopmans}} | \varphi_i \rangle = \left. \frac{dE^{\text{Koopmans}}}{df_i} \right|_{f_i=s}$$



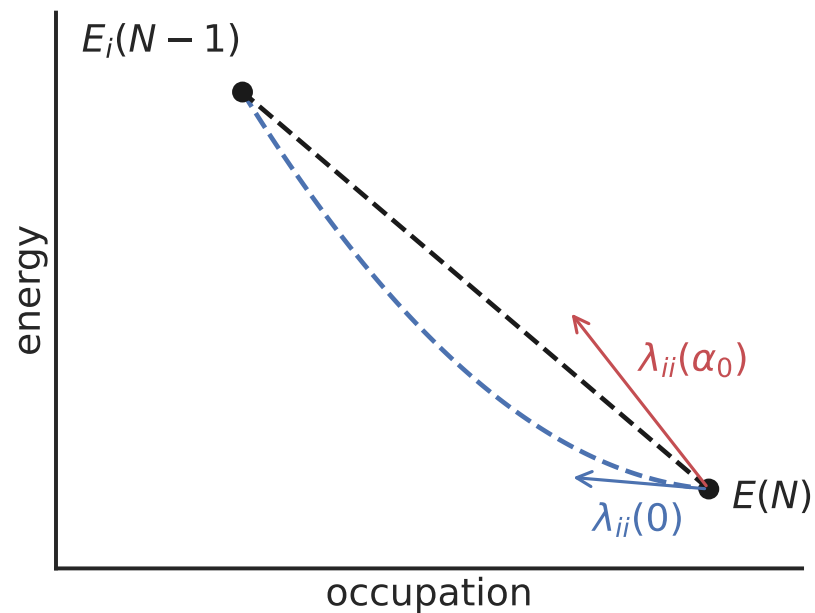
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Given this, how to work out the ideal α ?



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$$\alpha_{n+1} = \alpha_n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha_n) - \lambda_{ii}(0)}$$

$$\alpha^{n+1} = \alpha^n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha^n) - \lambda_{ii}(0)}$$

total energy with electron removed from orbital i

total energy of neutral system

expectation value of $\hat{H}^{\text{Koopmans}}$

expectation value of $\hat{H}^{\text{DFT}}$

Screening via DFPT

How can we avoid explicit charged defect calculations in a supercell?

¹N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

²N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022)

³R. De Gennaro *et al.* *Phys. Rev. B* **106**, 35106 (2022)

Screening via DFPT

How can we avoid explicit charged defect calculations in a supercell?

Reformulate in terms of DFPT¹...

$$\alpha_i = 1 + \frac{\langle v_{\text{pert}}^i | \Delta^i n \rangle}{\langle n_i | v_{\text{pert}}^i \rangle}$$

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... in reciprocal space²

$$\alpha_{0i} = 1 + \frac{\sum_{\mathbf{q}} \langle v_{\text{pert},\mathbf{q}}^{0i} | \Delta_{\mathbf{q}}^{0i} n \rangle}{\sum_{\mathbf{q}} \langle n_{\mathbf{q}}^{0i} | v_{\text{pert},\mathbf{q}}^{0i} \rangle}$$

(See Nicola Colonna's talk)

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(See Nicola Colonna's talk)

N.B. even for the supercell, we can still reconstruct a band structure³

¹N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

²N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022)

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To summarise...

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When performing a Koopmans calculation, you must decide...

- which flavour (KI, KIPZ)?
- how are we treating the orbital-density-dependence?
 - how are we initialising our variational orbitals? (N.B. depends on the flavour!)
 - are we going to explicitly minimise the ODD?
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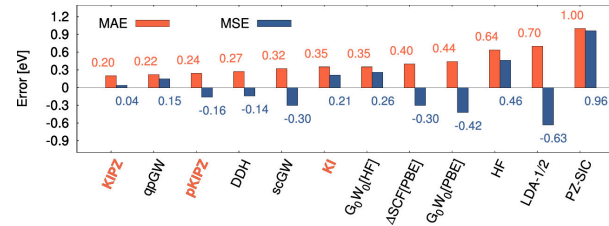
To summarise...

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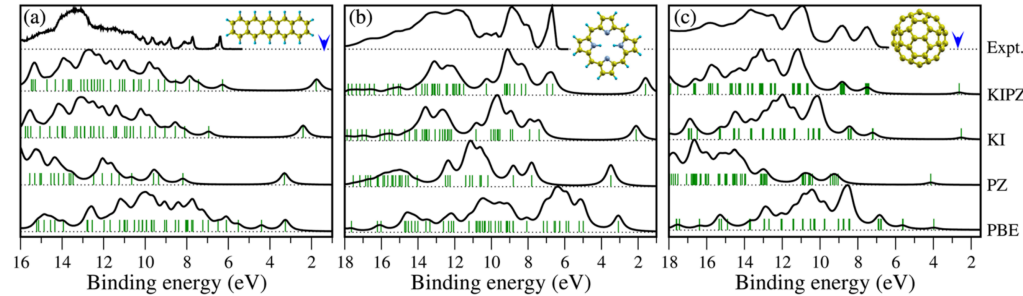
- which flavour (KI, KIPZ)?
- how are we treating the orbital-density-dependence?
 - how are we initialising our variational orbitals? (N.B. depends on the flavour!)
 - are we going to explicitly minimise the ODD?
- how are we calculating the screening parameters? (finite differences, DFPT, ML...)

Koopmans functionals: results for molecules

Ionisation potentials of 100 molecules cf. CCSD(T)



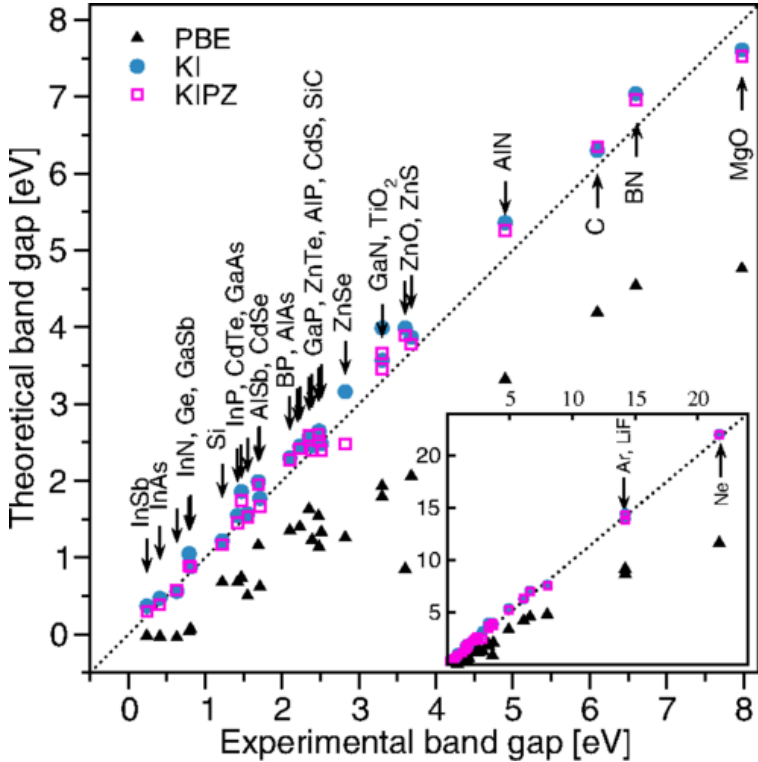
Ultraviolet photoemission spectra



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¹N. Colonna *et al.* *J. Chem. Theory Comput.* **14**, 2549–2557 (2018), N. L. Nguyen *et al.* *Phys. Rev. Lett.* **114**, 166405 (2015)

Koopmans functionals: results for solids



Mean absolute error (eV) across prototypical semiconductors and insulators

	PBE	G_0W_0	KI	KIPZ	$QSGW^{\sim}$
E_{gap}	2.54	0.56	0.27	0.22	0.18
IP	1.09	0.39	0.19	0.21	0.49

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¹N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018)

Koopmans functionals: results for solids

	PBE	G ₀ W ₀ ^a	scG $\widetilde{W}$ ^b	KI@[PBE,MLWFs]	KIPZ@PBE	exp ^c
E_g	0.49	1.06	1.14	1.16	1.15	1.17
$\Gamma_{1v} \rightarrow \Gamma_{25'v}$	11.97	12.04		11.97	12.09	12.5 ± 0.6
$X_{1v} \rightarrow \Gamma_{25'v}$	7.82			7.82		7.75
$X_{4v} \rightarrow \Gamma_{25'v}$	2.85	2.99		2.85	2.86	2.90
$L_{2'v} \rightarrow \Gamma_{25'v}$	9.63	9.79		9.63	9.74	9.3 ± 0.4
$L_{1v} \rightarrow \Gamma_{25'v}$	6.98	7.18		6.98	7.04	6.8 ± 0.2
$L_{3'v} \rightarrow \Gamma_{25'v}$	1.19	1.27		1.19		1.2 ± 0.2
$\Gamma_{25'v} \rightarrow \Gamma_{15c}$	2.48	3.29		3.17	3.20	3.35 ± 0.01
$\Gamma_{25'v} \rightarrow \Gamma_{2'c}$	3.28	4.02		3.95	3.95	4.15 ± 0.05
$\Gamma_{25'v} \rightarrow X_{1c}$	0.62	1.38		1.28	1.31	1.13
$\Gamma_{25'v} \rightarrow L_{1c}$	1.45	2.21		2.12	2.13	2.04 ± 0.06
$\Gamma_{25'v} \rightarrow L_{3c}$	3.24	4.18		3.91	3.94	3.9 ± 0.1
MSE	0.35	0.02		0.01	0.03	
MAE	0.44	0.21		0.14	0.17	

^{a1} for E_g and ^{a2} for the transitions; ^{b3}; ^{c4}

¹M. Shishkin *et al. Phys. Rev. B* **75**, 235102 (2007)

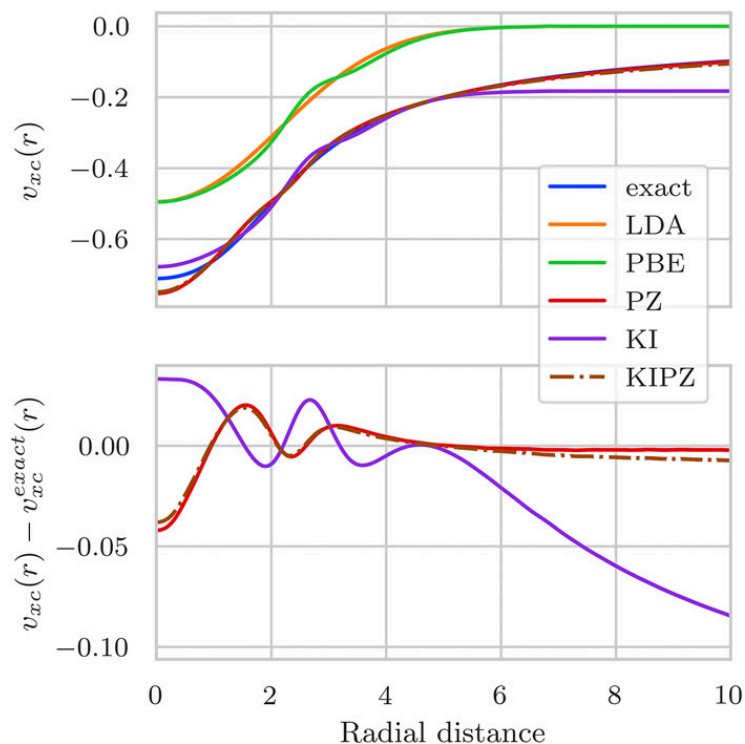
²M. S. Hybertsen *et al. Phys. Rev. B* **34**, 5390–5413 (1986)

³M. Shishkin *et al. Phys. Rev. Lett.* **99**, 246403 (2007)

⁴O. Madelung. (Springer-Verlag, Berlin, 2004).

Koopmans functionals: results for toy systems

For Hooke's atom (two electrons in a harmonic confining potential with Coulombic repulsion)



1

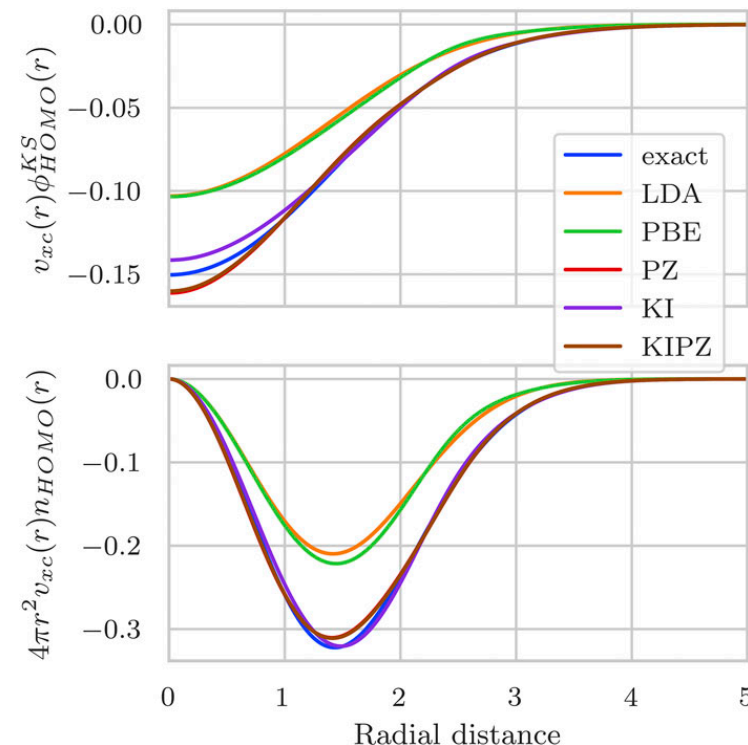
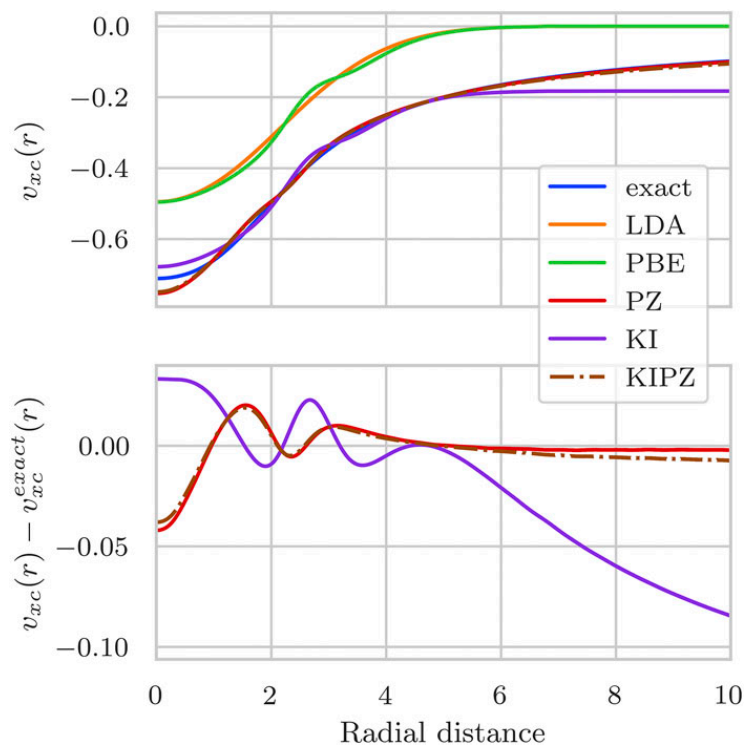
¹Y. Schubert *et al. J. Chem. Phys.* **158**, 144113 (2023)

Koopmans functionals: results for toy



systems

For Hooke's atom (two electrons in a harmonic confining potential with Coulombic repulsion)



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¹Y. Schubert *et al. J. Chem. Phys.* **158**, 144113 (2023)

Koopmans functionals: caveats

-

¹N. Q. Su *et al.* *J. Phys. Chem. Lett.* **11**, 1528–1535 (2020)

Koopmans functionals: caveats

- restricted to systems with a non-zero band gap
-

¹N. Q. Su *et al.* *J. Phys. Chem. Lett.* **11**, 1528–1535 (2020)

Koopmans functionals: caveats

- restricted to systems with a non-zero band gap
- empty state localization in the bulk limit
-

¹N. Q. Su *et al.* *J. Phys. Chem. Lett.* **11**, 1528–1535 (2020)

Koopmans functionals: caveats

- restricted to systems with a non-zero band gap
- empty state localization in the bulk limit
- can potentially break the crystal point group symmetry¹

¹N. Q. Su *et al.* *J. Phys. Chem. Lett.* **11**, 1528–1535 (2020)

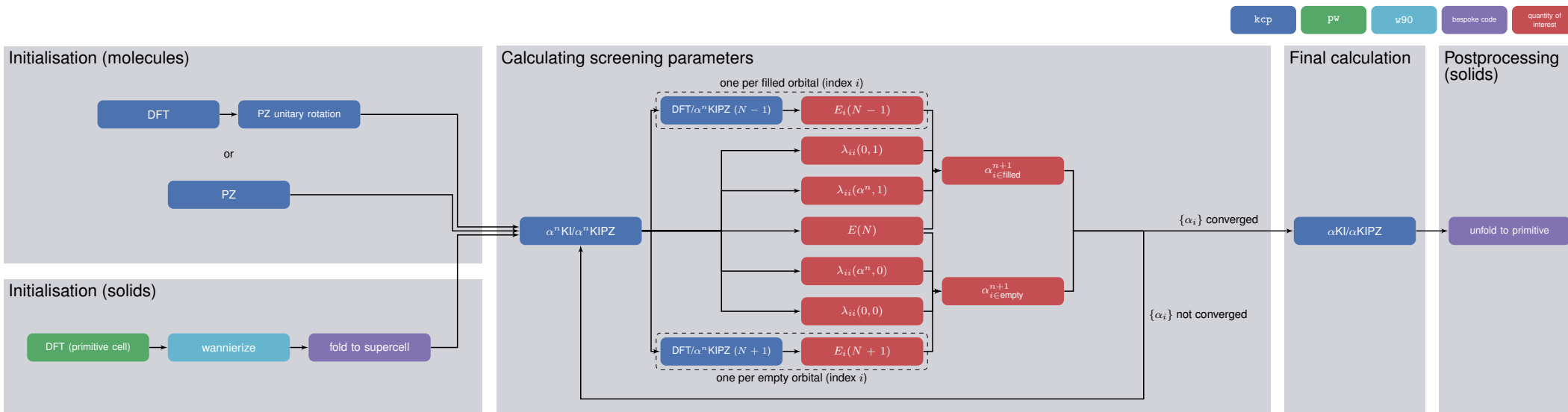
The workflows

The general workflow:

- define/initialize a set of variational orbitals
- calculate the screening parameters $\{\alpha_i\}$
- construct and diagonalize the Hamiltonian

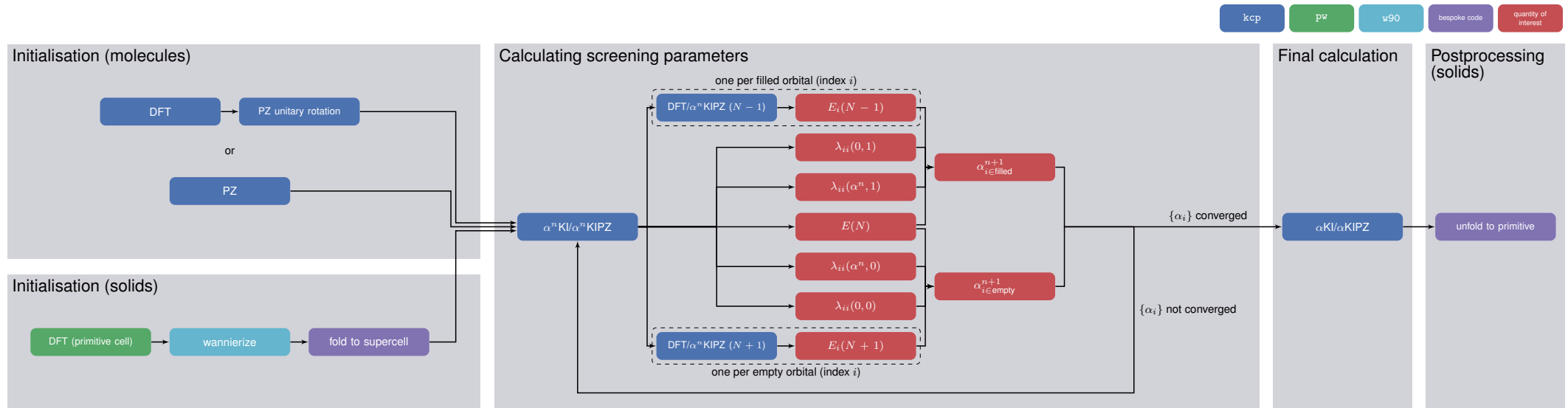
The workflows

(a) finite difference calculations using a supercell

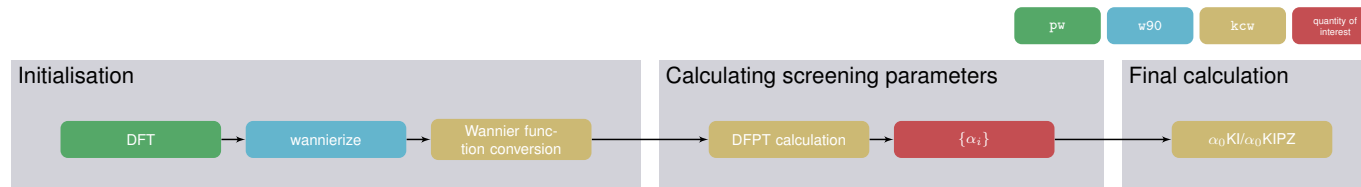


The workflows

(a) finite difference calculations using a supercell



(b) DFPT in a primitive cell



How do I run these calculations?

Complicated workflows mean that...

-

How do I run these calculations?

Complicated workflows mean that...

- lots of different codes that need to handshake
-

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How do I run these calculations?

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Our solution...



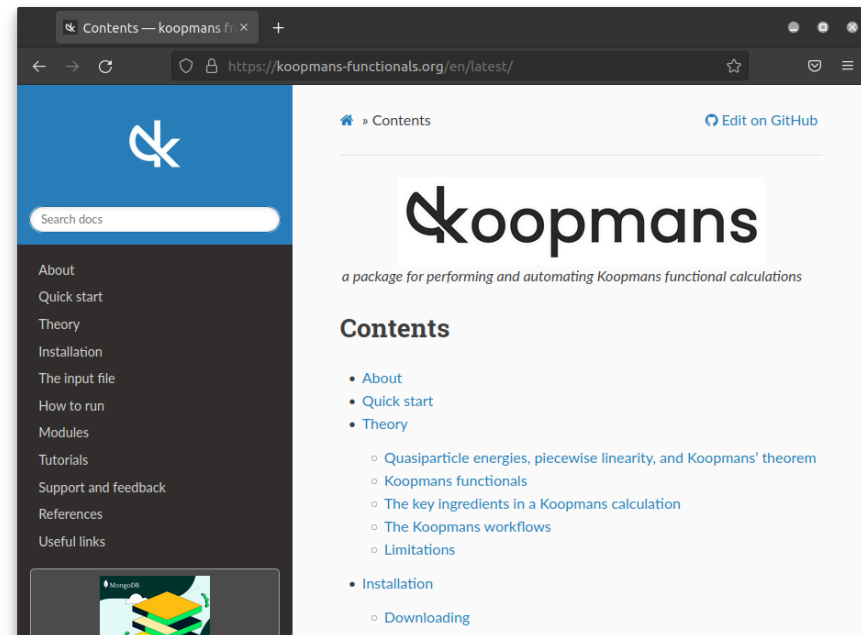
- v1.0 released earlier this year¹
- implementations of Koopmans functionals
- automated workflows
 - start-to-finish Koopmans calculations
 - Wannierisation
 - dielectric tensor
 - ...
- built on top of ASE²
- under the hood, calls Quantum ESPRESSO
- does not require expert knowledge

koopmans-functionals.org

¹E. B. Linscott *et al.* *J. Chem. Theory Comput.* **19**, 7097–7111 (2023)

²A. H. Larsen *et al.* *J. Phys. Condens. Matter* **29**, 273002 (2017)

The koopmans package



koopmans: the input file



```
{
  "workflow": {
    "task": "singlepoint",
    "functional": "ki",
    "method": "dscf",
    "init_orbitals": "mlwfs",
    "alpha_guess": 0.1
  },
  "atoms": {
    "atomic_positions": {
      "units": "crystal",
      "positions": [[{"Si", 0.00, 0.00, 0.00},
                     {"Si", 0.25, 0.25, 0.25}]
    ]
  },
  "cell_parameters": {
    "periodic": true,
    "ibrav": 2,
    "celldm(1)": 10.262
  }
},
```

```
  "k_points": {
    "grid": [8, 8, 8],
    "path": "LGXKG"
  },
  "calculator_parameters": {
    "ecutwfc": 60.0,
    "w90": {
      "projections": [
        [{"fsite": [0.125, 0.125, 0.125],
          "ang_mtm": "sp3"}],
        [{"fsite": [0.125, 0.125, 0.125],
          "ang_mtm": "sp3"}]
      ],
      "dis_froz_max": 11.5,
      "dis_win_max": 17.0
    }
  }
}
```

koopmans is scriptable



```
from ase.build import bulk
from koopmans.kpoints import Kpoints
from koopmans.projections import ProjectionBlocks
from koopmans.workflows import SinglepointWorkflow

# Use ASE to create bulk silicon
atoms = bulk('Si')

# Define the projections for the Wannierization (same for filled and empty manifold)
si_proj = [{'fsite': [0.25, 0.25, 0.25], 'ang_mtm': 'sp3'}]
si_projs = ProjectionBlocks.from_list([si_proj, si_proj], atoms=atoms)

# Create the workflow
workflow = SinglepointWorkflow(atoms = atoms,
                               projections = si_projs,
                               ecutwfc = 40.0,
                               kpoints = Kpoints(grid=[8, 8, 8], path='LGXKG', cell=atoms.cell),
                               calculator_parameters = {'pw': {'nbnd': 10},
                                                         'w90': {'dis_froz_max': 10.6, 'dis_win_max': 16.9}})

# Run the workflow
workflow.run()
```


Generic structure

Workflow

Generic structure

Workflow

atoms an ASE Atoms object

Generic structure

Workflow

atoms an ASE Atoms object

calculations a list of ASE calculators

Generic structure

Workflow

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kpoints a custom class containing k -point information

Generic structure

Workflow

atoms an ASE Atoms object

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Generic structure

Workflow

atoms an ASE Atoms object

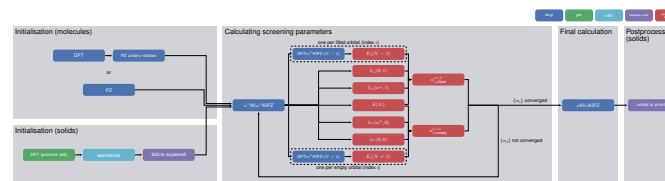
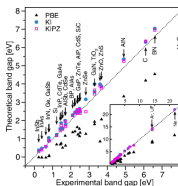
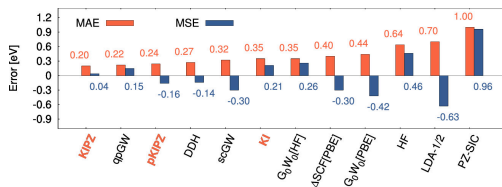
calculations a list of ASE calculators

kpoints a custom class containing k -point information

pseudopotentials a dictionary of pseudopotentials

We will see examples in the hands-on!

Take home messages



- Koopmans functionals are more complicated than a simple semi-local DFT calculation, because of...
 - orbital-density-dependence
 - screening parameters
- Koopmans functionals are implemented in Quantum ESPRESSO
- the complexity of the workflows are handled by the `koopmans` package

koopmans: An Open-Source Package for Accurately and Efficiently Predicting Spectral Properties with Koopmans Functionals

Edward B. Linscott,^{*,△} Nicola Colonna,[△] Riccardo De Gennaro, Ngoc Linh Nguyen, Giovanni Borghi, Andrea Ferretti, Ismaila Dabo, and Nicola Marzari^{*}



Cite This: <https://doi.org/10.1021/acs.jctc.3c00652>



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Metrics & More

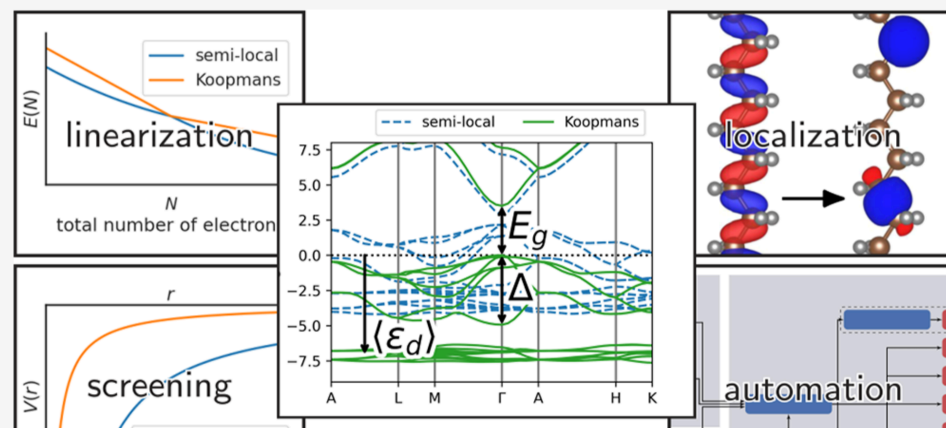


Article Recommendations



Supporting Information

ABSTRACT: Over the past decade we have developed Koopmans functionals, a computationally efficient approach for predicting spectral properties with an orbital-density-dependent functional framework. These functionals impose a generalized piecewise linearity condition to the entire electronic manifold, ensuring that orbital energies match the corresponding electron removal/addition energy differences (in contrast to semilocal DFT, where a mismatch between the two lies at the heart of the band gap problem and, more generally, the unreliability of Kohn–Sham orbital energies). This strategy has proven to be very powerful.



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Nicola Marzari



Nicola Colonna



Riccardo De Gennaro



Yannick Schubert



**Swiss National
Science Foundation**



Want to find out more? Go to koopmans-functionals.org

Follow  [@ed_linscott](https://twitter.com/ed_linscott) for updates | Slides available at  [github/elinscott](https://github.com/elinscott)

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Spare slides

Koopmans functionals: off-diagonal occupancies



Recap from earlier

Key idea: construct a functional such that the *variational* orbital energies

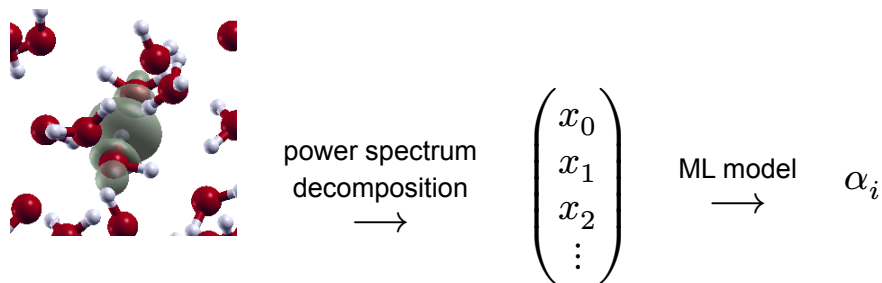
$$\varepsilon_i^{\text{Koopmans}} = \langle \varphi_i | H | \varphi_i \rangle = \partial E_{\text{Koopmans}} / \partial f_i$$

are...

- independent of the corresponding occupancies f_i
- equal to the corresponding total energy difference $E_i(N-1) - E(N)$

zero band gap $\rightarrow$ occupancy matrix for variational orbitals is off-diagonal

Learning the screening parameters



$$\rho_i(\mathbf{r})$$

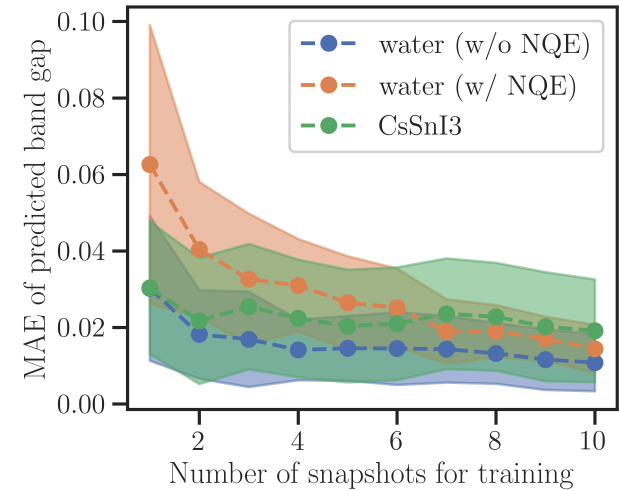
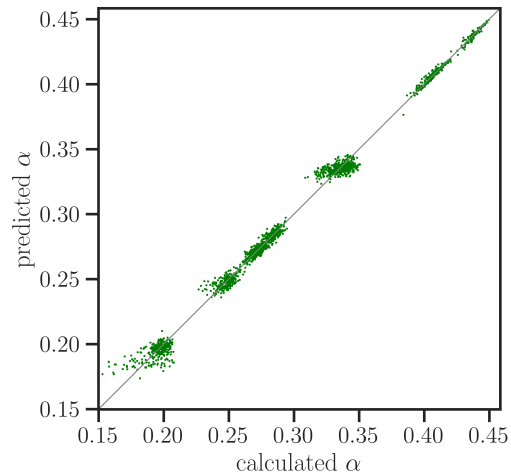
1

$$c_{nlm,k=\text{orbital}}^i = \int d\mathbf{r} g_{nl}(r) Y_{lm}(\theta, \varphi) \rho^i(\mathbf{r} - \mathbf{R}^i)$$

$$p_{n_1 n_2 l, k_1 k_2}^i = \pi \sqrt{\frac{8}{2l+1}} \sum_m c_{n_1 l m, k_1}^{i*} c_{n_2 l m, k_2}^i$$

¹Y. Schubert. (2022).

Resonance with other efforts



loss of accuracy of the band gap of ~ 0.02 eV

(cf. when calculating screening parameters *ab initio*)

speedup of $70 \times$

1

¹Y. Schubert. (2022).

Resonance with other efforts



- Wannier transition-state method of Anisimov and Kozhevnikov¹
- Optimally tuned hybrid functionals of Kronik, Pasquarello, and others (refer back to Leeor's talk on Wednesday)²
- Ensemble DFT of Kronik and co-workers³
- Koopmans-Wannier of Wang and co-workers⁴
- Dielectric-dependent hybrid functionals of Galli and co-workers⁵
- LOSC functionals of Yang and co-workers⁶

¹V. I. Anisimov *et al.* *Phys. Rev. B* **72**, 75125 (2005)

²L. Kronik *et al.* *J. Chem. Theory Comput.* **8**, 1515–1531 (2012), D. Wing *et al.* *Proc. Natl. Acad. Sci.* **118**, e2104556118 (2021)

³E. Kraisler *et al.* *Phys. Rev. Lett.* **110**, 126403 (2013)

⁴J. Ma *et al.* *Sci. Rep.* **6**, 24924 (2016)

⁵J. H. Skone *et al.* *Phys. Rev. B* **93**, 235106 (2016)

⁶C. Li *et al.* *Natl. Sci. Rev.* **5**, 203–215 (2018)